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***Abstract*—As autonomous driving technology advances, autonomous vehicles are expected not only to navigate complex traffic environments safely and efficiently but also to understand and respond to passengers' personalized service requirements. However, despite their strong semantic understanding capabilities, existing large language model (LLM)-based decision-making methods struggle to achieve both real-time responsiveness and physical executability. Conventional game-theoretic approaches offer advantages in computational efficiency and interpretability, but their representations of intentions and driving types typically rely on manually specified parameters. To address these limitations, this paper proposes an intent-conditioned fast–slow decision-making framework that integrates LLM reasoning with Bayesian games, combining their complementary strengths to unify intention understanding and reliable control. First, a scene encoder and key-agent selection module transform vehicle states, road topology, and passenger instructions into structured descriptions, reducing the reasoning burden of the LLM. Second, an intention reasoning module jointly infers passenger driving preferences and the latent behavioral types of surrounding vehicles. The inferred preferences and type probabilities are mapped to multi-objective payoff weights and Bayesian beliefs, respectively. Finally, a parameterized Bayesian–Stackelberg game jointly optimizes discrete lateral maneuvers and continuous longitudinal accelerations under incomplete information, while model-based trajectory generation and control ensure dynamically feasible execution. The framework is evaluated in HighwayEnv across cruising, ramp merging, blocking, cut-in, and instruction-change scenarios. Comparative and ablation results demonstrate improved task success and traffic efficiency while maintaining safety and ride comfort, and show that intention-conditioned beliefs enable proactive and personalized driving behavior.**

***Index Terms*—Autonomous driving, Bayesian game, driving intention, fast–slow decision-making, large language model, personalized driving, Stackelberg game.**

I. Introduction

RECENT advances in autonomous driving have substantially improved road safety, traffic efficiency, and mobility services [**?**]. As driving automation progresses toward higher levels, however, the human role gradually shifts from an active driver to a passenger. Autonomous vehicles must therefore not only complete driving tasks safely, but also understand passenger requests expressed in natural language and adjust their behavior accordingly. For instance, an urgent trip may prioritize efficiency, whereas a comfort-oriented request may favor smoother acceleration and larger safety margins. At the same time, the vehicle must interact with surrounding traffic participants whose intentions and responses are only partially observable. These requirements call for a decision-making system that connects semantic understanding, interaction reasoning, and dynamically feasible execution.

Existing decision-making methods can generally be divided into rule-based, learning-based, and game-theoretic approaches. Rule-based methods are efficient and interpretable, but rely heavily on handcrafted rules and predefined states, which limits their scalability in complex scenarios [**?**]. Learning-based methods, particularly deep reinforcement learning, can learn flexible policies from interaction data, yet often exhibit limited generalization and interpretability. Moreover, conventional prediction–planning pipelines typically treat predicted motions as fixed and ignore the influence of the ego vehicle's actions on other agents, which may lead to conservative or frozen behavior [**?**]. Game theory offers a principled alternative by explicitly modeling such coupled responses. Stackelberg games, mixed-strategy Nash equilibria, learned payoff functions, stochastic games, and Bayesian games have been applied to lane changing, merging, and intersection decision-making [**?**], [**?**], [**?**], [**?**], [**?**]. Nevertheless, most of these methods require driving types, prior beliefs, and payoff weights to be manually specified or estimated from predefined motion models, making it difficult to accommodate open-ended passenger requests and rapidly changing surrounding-vehicle intentions.

Recent large language models (LLMs) provide new possibilities for addressing these limitations through their world knowledge and natural-language reasoning capabilities [**?**]. Existing studies have used LLMs for knowledge-driven reflection, passenger-instruction interpretation, language-based planning, and cooperative decision-making [**?**], [**?**], [**?**], [**?**], [**?**]. Although these studies demonstrate improved semantic understanding and human–vehicle interaction, directly deploying an LLM in a real-time control loop remains impractical. LLM outputs may be temporally inconsistent or physically infeasible, and their inference latency cannot satisfy the frequency required for vehicle control. Hierarchical and fast–slow architectures address executability by combining semantic reasoning with model-based controllers or learned policies [**?**], [**?**], [**?**], but their interfaces are commonly limited to discrete commands. Consequently, semantic uncertainty is rarely propagated into downstream optimization, and passenger preferences are not explicitly converted into interpretable objective parameters.

Passenger and surrounding-vehicle intentions serve different but complementary roles in interactive driving. Passenger intentions determine how the ego vehicle should balance safety, efficiency, and ride comfort, whereas surrounding-vehicle intentions characterize uncertainty in the responses of other agents. Treating both merely as discrete semantic labels discards information that could directly guide optimization. A more effective interface is therefore to map passenger preferences to continuous payoff weights and represent surrounding-agent intentions as probability distributions over behavioral types. This formulation can exploit the semantic reasoning ability of an LLM while retaining the interpretability, constraint handling, and real-time execution capability of a model-based game. It also naturally supports a fast–slow structure in which semantic information is updated only when meaningful changes occur, while physical decisions are continuously recomputed using the latest preferences and beliefs.

To bridge this gap, this paper proposes a fast–slow autonomous driving decision-making framework that integrates LLM-based intention reasoning with a parameterized

Bayesian–Stackelberg game. The framework encodes vehicle states, road topology, motion histories, and passenger instructions into a structured scene description, and selects the agents most relevant to the ego vehicle. In the slow system, the LLM jointly infers passenger preferences and the latent behavioral types of selected surrounding vehicles. Passenger preferences are mapped to safety, efficiency, and comfort weights, while behavioral-type likelihoods form Bayesian beliefs over possible opponent intentions. In the fast system, these outputs parameterize a Bayesian–Stackelberg game that evaluates intention-dependent expected utilities and jointly determines discrete lateral maneuvers and continuous longitudinal accelerations under incomplete information. Model-based trajectory generation and control then ensure dynamically feasible and temporally smooth execution. Through asynchronous coupling and closed-loop belief updates, the framework adapts to semantic and behavioral changes without requiring the LLM to control the vehicle at every simulation step.

The main contributions of this paper are summarized as follows:

1) An intent-conditioned fast–slow architecture is proposed to connect low-frequency semantic reasoning with high-frequency vehicle decision-making and control. Structured scene encoding and key-agent selection enable the LLM to jointly reason about passenger requirements and surrounding-vehicle intentions while reducing reasoning and game-solving complexity.
2) A parameterized Bayesian–Stackelberg game is developed as an interpretable interface between the LLM and the physical decision layer. Passenger preferences are mapped to multi-objective payoff weights, while inferred surrounding-vehicle intentions are represented as Bayesian type beliefs for joint lateral and longitudinal decision-making.
3) A closed-loop evaluation framework is constructed on an extended HighwayEnv platform, covering highway cruising, ramp merging, adversarial blocking, sudden cut-in, and dynamic passenger-instruction scenarios. Comparative experiments, ablation studies, and representative cases assess task success, efficiency, safety, comfort, and personalized instruction compliance.

The remainder of this paper is organized as follows. Section II formulates the interactive driving problem. Section III presents the proposed framework. Section IV describes the experimental setup, and Section V reports the results. Finally, Section VI concludes this paper.

II. Problem Formulation

A. Interactive Environment and Intent Representation

Consider an autonomous ego vehicle, indexed by 0, interacting with the observable surrounding vehicles $\mathcal{N}_k = \{1, \dots, N_k\}$ at decision step k. The state of vehicle $i \in \{0\} \cup \mathcal{N}_k$ is

$$\mathbf{x}_{i,k} = \left[p^x_{i,k}, p^y_{i,k}, v^x_{i,k}, v^y_{i,k}, \psi_{i,k}, a_{i,k}, \ell_{i,k}\right]^\top, \tag{1}$$

where the elements denote position, velocity, heading angle, longitudinal acceleration, and occupied lane. Given the local road context $\mathcal{M}_k$, the joint traffic state is

$$\mathbf{s}_k = (\mathcal{M}_k, \mathbf{x}_{0,k}, \{\mathbf{x}_{i,k}\}_{i\in\mathcal{N}_k}) \in \mathcal{S}. \tag{2}$$

Because the complete state is unavailable under limited perception, occlusion, and measurement noise, the ego vehicle receives $\mathbf{o}_k = h(\mathbf{s}_k) + \boldsymbol{\nu}_k$ and maintains the finite observation history

$$\mathcal{H}_k = \{\mathbf{o}_{k-H_\mathrm{o}+1}, \dots, \mathbf{o}_k\}, \tag{3}$$

which characterizes the recent motion of surrounding vehicles. Only the key vehicles $\mathcal{K}_k \subseteq \mathcal{N}_k$ that may affect the ego vehicle within the prediction horizon are included in interaction reasoning; their selection is described in Section III. The problem is therefore partially observable and interactive: the future ego state depends on both its own action and the unknown responses of these key vehicles.

Two forms of intention are considered. At slow reasoning step m, a passenger provides a natural-language instruction q_m, whose underlying preference is represented by

$$\boldsymbol{\omega}_m = [\omega^\mathrm{s}_m, \omega^\mathrm{e}_m, \omega^\mathrm{c}_m]^\top, \quad \omega^r_m \geq 0, \quad \sum_{r\in\{\mathrm{s,e,c}\}} \omega^r_m = 1, \tag{4}$$

where the components weight safety, efficiency, and comfort. In contrast, the behavioral type $\theta_{i,k} \in \Theta_i$ of each key vehicle is latent. The ego vehicle therefore maintains the belief

$$b_{i,k}(\theta) = \Pr(\theta_{i,k} = \theta \mid \mathcal{H}_k, \mathcal{M}_k), \qquad \sum_{\theta\in\Theta_i} b_{i,k}(\theta) = 1. \tag{5}$$

Let $\boldsymbol{\theta}_k = [\theta_{i,k}]_{i\in\mathcal{K}_k}$ denote the joint type vector. Assuming conditional independence given the observed scene, its belief distribution is

$$b_k(\boldsymbol{\theta}) = \prod_{i\in\mathcal{K}_k} b_{i,k}(\theta_i). \tag{6}$$

Each type remains fixed within one short game horizon but may change between horizons, enabling closed-loop inference from newly observed motion. Passenger preferences condition the ego utility, whereas type beliefs represent uncertainty in surrounding-vehicle responses.

B. Intent-Conditioned Bayesian–Stackelberg Decision

Each participant selects a hybrid action

$$\mathbf{u}_{i,k} = (d_{i,k}, a^\mathrm{cmd}_{i,k}), \quad d_{i,k} \in \mathcal{D}_{i,k}, \quad a^\mathrm{min}_{i,k} \leq a^\mathrm{cmd}_{i,k} \leq a^\mathrm{max}_{i,k}, \tag{7}$$

where $\mathcal{D}_{i,k}$ contains the feasible left-change, lane-keeping, and right-change maneuvers; actions toward nonexistent lanes are excluded. Over a game horizon H_g, the action sequence is

$$\mathbf{U}_{i,k} = \{\mathbf{u}_{i,k|k}, \dots, \mathbf{u}_{i,k+H_\mathrm{g}-1|k}\}. \tag{8}$$

A model-based generator maps this sequence to a trajectory satisfying road, kinematic, actuation, comfort, and collision-avoidance constraints. The joint state evolves as

$$\mathbf{s}_{k+1} = F(\mathbf{s}_k, \mathbf{u}_{0,k}, \mathbf{u}_{-0,k}), \tag{9}$$

where $\mathbf{u}_{-0,k}$ denotes the actions of the key surrounding vehicles.

The ego vehicle is modeled as the Stackelberg leader and the key vehicles as followers with unknown types. For an ego sequence $\mathbf{U}_{0,k}$ and a type realization $\boldsymbol{\theta}$, their type-dependent equilibrium response is

$$\mathbf{U}^*_{-0,k}(\boldsymbol{\theta}; \mathbf{U}_{0,k}) \in \mathcal{R}_k(\boldsymbol{\theta}, \mathbf{U}_{0,k}), \tag{10}$$

where $\mathcal{R}_k$ is induced by the followers' type-dependent utilities. Thus, yielding and competing vehicles may respond differently to the same ego maneuver. Let $\mathbf{r}_0 = [r_0^{\mathrm{s}}, r_0^{\mathrm{e}}, r_0^{\mathrm{c}}]^\top$ collect the safety, efficiency, and comfort performance. The ego utility is

$$J_0 = \sum_{\tau=0}^{H_g-1} \gamma^\tau \boldsymbol{\omega}_m^\top \mathbf{r}_0(\mathbf{s}_{k+\tau}, \mathbf{u}_{0,k+\tau}, \mathbf{u}_{-0,k+\tau}). \tag{11}$$

Accordingly, the ego decision is formulated as

$$\begin{aligned} \mathbf{U}^*_{0,k} \in \arg\max_{\mathbf{U}_{0,k} \in \mathcal{F}_{0,k}} \quad & \sum_{\boldsymbol{\theta} \in \boldsymbol{\Theta}_k} b_k(\boldsymbol{\theta}) J_0\big(\mathbf{s}_k, \mathbf{U}_{0,k}, \mathbf{U}^*_{-0,k}(\boldsymbol{\theta}; \mathbf{U}_{0,k}); \boldsymbol{\omega}_m\big) \\ \text{s.t.} \quad & \mathbf{U}^*_{-0,k}(\boldsymbol{\theta}; \mathbf{U}_{0,k}) \in \mathcal{R}_k(\boldsymbol{\theta}, \mathbf{U}_{0,k}), \\ & \mathbf{s}_{k+\tau+1} = F(\mathbf{s}_{k+\tau}, \mathbf{u}_{0,k+\tau}, \mathbf{u}_{-0,k+\tau}), \\ & \boldsymbol{\tau}_{0,k} \in \mathcal{F}_{0,k}. \end{aligned} \tag{12}$$

Thus, the ego vehicle maximizes its expected intent-conditioned utility over possible surrounding-vehicle types rather than optimizing against a single deterministic prediction.

C. Fast–Slow Decision Requirement

LLM reasoning need not operate at the vehicle-control frequency. We distinguish the LLM, game-solving, and control intervals as

$$\Delta t_{\mathrm{L}} \gg \Delta t_{\mathrm{G}} \geq \Delta t_{\mathrm{C}}. \tag{13}$$

The slow system updates $\boldsymbol{\omega}_m$ and the intention priors periodically or when a semantic or interaction event occurs. Between two slow updates, the preference is retained and the beliefs are recursively refined from newly observed motions according to

$$b_{i,k+1} = \mathcal{B}_i(b_{i,k}, \mathbf{o}_{k+1}, \mathbf{u}_{0,k}), \tag{14}$$

where $\mathcal{B}_i$ denotes the Bayesian update operator. The fast layer repeatedly solves (12), while the control layer tracks the resulting trajectory. The overall problem is therefore to construct

$$\Pi : (q_m, \mathcal{H}_k, \mathcal{M}_k) \mapsto (\boldsymbol{\omega}_m, b_k) \mapsto \mathbf{U}^*_{0,k} \mapsto \boldsymbol{\tau}^*_{0,k}, \tag{15}$$

such that the executed behavior follows the passenger requirement, accounts for uncertain interactions, satisfies safety and vehicle-feasibility constraints, and remains operational when the slow semantic module is delayed.

III. Intent-Conditioned Fast–Slow Cooperative Decision-Making

A. Overall Architecture

To combine the semantic reasoning capability of large language models (LLMs) with the real-time requirements of vehicle control, we propose the intent-conditioned fast–slow framework shown in Fig. 1. The framework consists of scene preprocessing, asynchronous dual-intention reasoning, parameter mapping, candidate-path generation, interaction-aware game solving, and trajectory tracking. The slow system interprets passenger instructions and estimates probabilistic intentions of surrounding vehicles, whereas the fast system generates executable actions under road, vehicle, and safety constraints. The LLM does not directly output steering or acceleration commands; instead, its semantic outputs are converted into bounded model parameters and Bayesian beliefs.

At each planning step, the ego state, road topology, and predicted trajectories of surrounding vehicles are first encoded into a compact scene description. An independent LLM thread returns a passenger-style label, parameter adjustments, and type probabilities of selected surrounding vehicles. The latest valid outputs parameterize the desired speed, lane-change geometry, safety thresholds, utility weights, and type-conditioned response model without blocking the fast loop. For every feasible lane, the fast system generates lane-keeping, partially probing, and complete lane-change paths, solves the longitudinal interaction along each path, and selects the branch with the highest intent-conditioned utility. A Stanley controller then tracks the selected reference trajectory.

B. Interaction Object Selection and Scene Encoding

Let the ego state at step k be

$$\mathbf{x}_{0,k} = [x_{0,k}, y_{0,k}, \psi_{0,k}, v_{0,k}, a_{0,k}, \delta_{0,k}]^\top. \tag{16}$$

To limit the game size while preserving critical interactions, surrounding vehicles are selected in two stages. First, lane-topology filtering retains the nearest preceding vehicle in the current lane and the nearest preceding and following vehicles in each adjacent candidate lane. Vehicle centers and corners are jointly projected onto lane centerlines to stabilize selection during lane crossing. Second, if the number of selected vehicles remains below the model capacity, vehicles whose predicted trajectories approach the ego path within a 1–8 s horizon are added according to their first hazardous-entry time and minimum predicted distance. The resulting set is

$$\begin{aligned} \mathcal{K}_k^{\mathrm{game}} = \mathrm{Trunc}_{N_g}\left(\mathcal{K}_k^{\mathrm{lane}} \cup \mathcal{K}_k^{\mathrm{spatiotemporal}}\right), \\ N_g = 5. \end{aligned} \tag{17}$$

This topology-first strategy preserves explicit traffic relationships and bounds the computational scale without introducing a manually weighted relevance score.

The selected vehicles are represented in a Frenet frame defined by the current lane. For vehicle i,

$$\begin{aligned} &\Delta s_{i,k} = s_{i,k} - s_{0,k}, \quad \Delta l_{i,k} = l_{i,k} - l_{0,k}, \\ &\Delta v_{i,k} = v_{i,k} - v_{0,k}. \end{aligned} \tag{18}$$

The slow system maintains a five-frame history

$$\mathcal{H}_k = \{\mathbf{x}_{0,k-h}, \mathbf{o}_{k-h}\}_{h=0}^{H_L-1}, \qquad H_L = 5, \tag{19}$$

where $\mathbf{o}_k$ contains vehicle identifiers, motion states, lane indices, and relative Frenet coordinates. The history maintains vehicle identity across asynchronous updates, while the latest relative states provide the primary numerical prompt inputs.

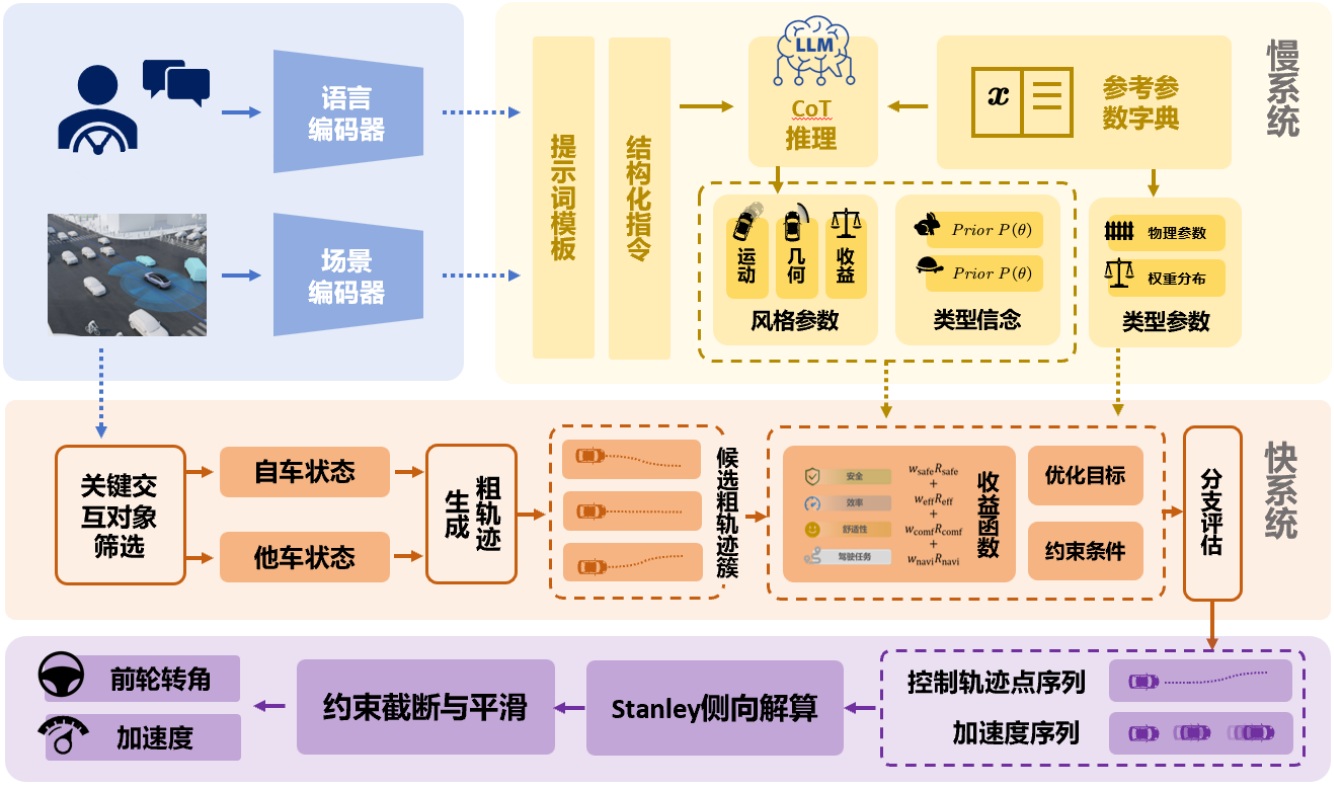


Fig. 1. Overall architecture of the proposed intent-conditioned fast–slow cooperative decision-making framework. The asynchronous slow system maps passenger instructions and structured traffic scenes to driving parameters and probabilistic type beliefs, while the model-based fast system performs path generation, interaction solving, branch selection, and trajectory tracking.

C. Structured Prompt and Asynchronous Intention Reasoning

To control prompt length, the slow system selects the three nearest vehicles from the game set:

$$\mathcal{K}_k^{\mathrm{LLM}} = \mathrm{TopK}_{i\in\mathcal{K}_k^{\mathrm{game}}}\left(-\sqrt{\Delta s_{i,k}^2+\Delta l_{i,k}^2},\,3\right). \tag{20}$$

Each target vehicle is encoded by $\mathbf{e}_{i,k} = [\mathrm{lane}(i,k), \Delta s_{i,k}, \Delta l_{i,k}, \Delta v_{i,k}]$, which exposes its relative lane, interaction geometry, and motion tendency without depending on global coordinates.

As illustrated in Fig. 2, a static instruction block is combined with a compact runtime block:

$$\mathcal{I}_k = \left[\mathcal{T}^{\mathrm{sys}}; \mathcal{T}^{\mathrm{run}}\left(u_k^{\mathrm{pass}}, v_{0,k}, a_{0,k}, \{\mathbf{e}_{i,k}\}_{i\in\mathcal{K}_k^{\mathrm{LLM}}}\right)\right]. \tag{21}$$

The static block specifies the autonomous-driving strategy role, the baseline style library, surrounding-type definitions, safety ranges, and JSON schema. The runtime block contains only the current passenger instruction, ego motion, and selected vehicle encodings.

The LLM performs two parallel reasoning tasks. First, it maps the passenger instruction to $z_k \in \{\mathrm{Aggressive}, \mathrm{Normal}, \mathrm{Conservative}\}$ and predicts bounded residuals around the corresponding baseline parameters:

$$\boldsymbol{\rho}_k = \mathrm{clip}\left(\bar{\boldsymbol{\rho}}(z_k) + \Delta\boldsymbol{\rho}_k, \boldsymbol{\rho}^{\min}, \boldsymbol{\rho}^{\max}\right). \tag{22}$$

The vector $\boldsymbol{\rho}_k$ includes desired-speed, lane-change, car-following, geometric-safety, and utility parameters. Second, for each selected surrounding vehicle, the LLM returns a belief over $\Theta = \{\mathrm{Yield}, \mathrm{Normal}, \mathrm{Force}\}$:

$$\mathbf{p}_{i,k} = [p_{i,k}^Y, p_{i,k}^N, p_{i,k}^F]^\top, \qquad \mathbf{1}^\top\mathbf{p}_{i,k} = 1. \tag{23}$$

Yield denotes a tendency to decelerate or enlarge the interaction gap, Normal mainly maintains the current motion, and Force prioritizes progress and road space.

The experimental system invokes `qwen-turbo` with a temperature of 0.4. Its JSON output is accepted only when all required fields are present, parameter values remain within their allowed ranges, and the vehicle-type scores can be normalized. Invalid vehicle outputs fall back to the normal type, whereas invalid passenger fields retain the latest valid parameter vector or, before the first valid response, the Normal profile.

D. Intent-to-Model Parameterization

The validated passenger output does not directly overwrite physical commands. For each style z, the reference dictionary stores a calibrated baseline $\bar{\boldsymbol{\rho}}^z$ and an admissible domain Ω_z. After applying (22), the resulting vector is written as

$$\boldsymbol{\rho}_k = [v_{0,k}^+, T_{\mathrm{lc},k}, d_{\mathrm{safe},k}, \kappa_{\mathrm{TTC},k}, w_{s,k}, w_{e,k}, w_{c,k}, \ldots]^\top, \tag{24}$$

where $v_{0,k}^+$ and $T_{\mathrm{lc},k}$ control motion efficiency and lane-change duration, $d_{\mathrm{safe},k}$ and $\kappa_{\mathrm{TTC},k}$ define geometric safety margins,

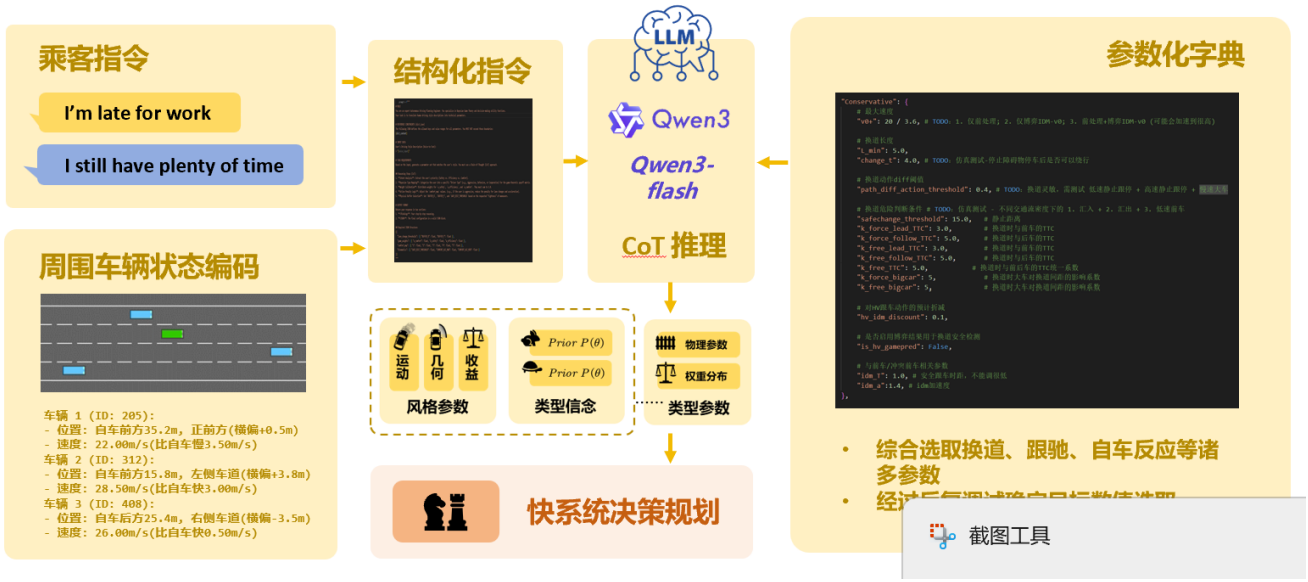


Fig. 2. Structured prompt construction for passenger-style adaptation and surrounding-vehicle intention inference. The LLM infers bounded style parameters and probabilistic behavioral types and returns a validated JSON interface for the fast decision layer.

and $[w_{s,k}, w_{e,k}, w_{c,k}]^\top$ weight safety, efficiency, and comfort. The desired speed is

$$v_{\text{des}} = \left(v_{\max}^{\text{AV}} + v_{0,k}^{+}\right) c_{v,k}, \tag{25}$$

where $c_{v,k}$ is the style-dependent speed coefficient. Thus, a passenger instruction can jointly affect motion, path geometry, safety filtering, and game utility while all parameters remain within prevalidated ranges.

The surrounding-vehicle beliefs parameterize a fixed family of response models. Yield emphasizes safety and active gap creation, Normal balances safety and efficiency while largely maintaining its motion, and Force assigns high priority to progress. For both ego and surrounding vehicles, the LLM can only select or adjust parameters within these verified model sets; it cannot generate arbitrary accelerations or bypass safety checks. This explicit interface improves interpretability and prevents numerical instability caused by unconstrained language outputs.

For controlled comparisons, an external preset mode directly loads $\bar{\boldsymbol{\rho}}^{z^\star}$ for a specified style. The online and preset modes therefore share the same parameter dictionary, candidate paths, game solver, and controller; they differ only in whether the style parameters originate from the passenger-instruction reasoner or an experimental setting.

E. Intent-Conditioned Candidate Path Generation

The fast system constructs a branch set $\mathcal{B}_k$ from the current lane, navigation candidate lanes, and lane-change state. The longitudinal acceleration of each branch is optimized separately, avoiding a mixed-integer nonlinear program while retaining continuous speed control.

For a target-lane branch, the lane-change length is

$$L_b = \text{clip}(v_{0,k} T_{\text{lc}}, L_{\min}, L_{\max}) \, \text{clip}\left(\frac{d_{\text{lane}}}{L_\Delta}, 0, 1\right), \tag{26}$$

where d_{lane} is the available longitudinal distance. A cubic Hermite curve

$$y_b(\ell) = c_3\ell^3 + c_2\ell^2 + c_1\ell + c_0, \qquad 0 \le \ell \le L_b, \tag{27}$$

is generated subject to

$$\begin{aligned} y_b(0) &= 0, & y_b'(0) &= 0, \\ y_b(L_b) &= r_b y_{\text{tar}}, & y_b'(L_b) &= \tan(r_b \Delta\psi_{\text{tar}}). \end{aligned} \tag{28}$$

The curve is sampled at 0.2 m intervals and connected to the target centerline to form Γ_b^{geo}.

Predicted surrounding trajectories are densified before geometric conflicts with each candidate path are located. Vehicle dimensions and relative headings are considered when computing conflict gaps. A branch is discarded if it violates the style-dependent gap and TTC thresholds, lacks sufficient prediction coverage or end-of-road braking distance, or intersects a predicted vehicle envelope.

F. Intent-Conditioned Longitudinal Interaction Game

Each feasible geometric path is paired with type-conditioned longitudinal rollouts. Along branch b, vehicle motion follows

$$s_{j+1} = s_j + v_j \Delta t_j + \frac{1}{2} a_j \Delta t_j^2, \qquad v_{j+1} = v_j + a_j \Delta t_j, \tag{29}$$

subject to $a_{\min} \le a_j \le a_{\max}$ and $0 \le v_j \le v_{\max}$.

For surrounding vehicle i of type θ, safety is evaluated from TTC and the arrival-time difference at the conflict point:

$$R_{\text{safe},i}^{b,\theta} = \text{smax}\left(\text{TTC}_i^{b,\theta}, \text{TTCP}_i^{b,\theta}\right), \tag{30}$$

where both quantities are clipped to $[0, 5]$ s. The expected safety of a branch is determined by its most critical interaction object:

$$R_{\text{safe}}^{b} = \min_{i \in \mathcal{K}_k^{\text{game}}} \sum_{\theta \in \Theta} p_{i,k}^{\theta} R_{\text{safe},i}^{b,\theta}. \tag{31}$$

Efficiency is measured by $R^b_{\text{eff},j} = v_j$, while comfort is $R^b_{\text{comf},j} = -\sqrt{(a_j - a_{j-1})^2 + \varepsilon}$. The ego acceleration is obtained from

$$\begin{aligned}\mathbf{a}_b^* = \arg\max_{\mathbf{a}_b} \quad & \sum_{j=1}^{N_r} e^{-\lambda t_j}\Big(w_{s,k} R^b_{\text{safe},j} \\ & + w_{e,k} R^b_{\text{eff},j} + w_{c,k} R^b_{\text{comf},j}\Big) \\ \text{s.t.} \quad & (29) \text{ and the physical bounds.}\end{aligned} \tag{32}$$

The weights are updated from the passenger parameters in (22) and validated before optimization. CasADi/IPOPT is used with $N_r = 1$ for utility evaluation, while state propagation and constraints cover all planning nodes.

For each surrounding vehicle and type, the follower searches a discrete acceleration set $\mathcal{A}_i^\theta$ using fixed type-dependent motion and utility parameters:

$$\begin{aligned}a_i^{*,\theta} = \arg\max_{a \in \mathcal{A}_i^\theta} \Big(& w_{s,\theta} R_{\text{safe},i} + w_{e,\theta} R_{\text{eff},i} \\ & + w_{c,\theta} R_{\text{comf},i}\Big).\end{aligned} \tag{33}$$

Yield, Normal, and Force emphasize safety and efficiency differently, thereby producing distinct responses to the same ego maneuver. The procedure forms a one-round Bayesian–Stackelberg approximation: the ego continuously optimizes as the leader, while type-conditioned followers perform discrete best-response searches. It avoids enumerating the joint type space and grows approximately linearly with the numbers of vehicles and types. The selected branch is

$$b_k^* = \arg\max_{b \in \mathcal{B}_k^{\text{feas}}} J_b^*. \tag{34}$$

G. Trajectory Construction and Control

The optimal accelerations are interpolated from the nonuniform planning nodes to an execution axis with a 0.0792 s interval and an approximately 8 s horizon. Integrating (29) yields the longitudinal path coordinate, which is mapped onto the selected geometric path to obtain

$$\Gamma_{0,k}^* = \{x_n, y_n, \psi_n, v_n, \tau_n\}_{n=0}^{N_o}. \tag{35}$$

Positions beyond the discrete path end are extrapolated along its terminal tangent.

The Stanley controller computes the raw front-wheel angle from lateral and heading errors:

$$\delta_k^{\text{raw}} = k_\psi e_{\psi,k} + \operatorname{atan2}\left(-k_e e_{y,k}, \max(v_{0,k}, 0.1)\right), \tag{36}$$

The steering angle is limited by mechanical, lateral-acceleration, and steering-rate constraints, while the current longitudinal command is taken from the optimal acceleration sequence. This closes the loop from passenger preferences and probabilistic intentions to physical vehicle actions.

H. Algorithm and Computational Complexity

The complete procedure is summarized in Algorithm 1.

Let B be the number of geometric branches, $N_g \le 5$ the number of game vehicles, $N_\theta = 3$ the number of types, N_a the maximum number of discrete actions, and C_{NLP} the cost of one ego nonlinear program. The dominant fast-loop complexity is

$$\mathcal{O}\left[B\left(C_{\text{NLP}} + N_g N_\theta N_a + C_{\text{conf}}\right)\right], \tag{37}$$

where C_{conf} is the conflict-detection cost. The method does not enumerate the $N_\theta^{N_g}$ joint type combinations. LLM inference is executed asynchronously and is therefore excluded from the blocking time of the fast planning loop; its information age is evaluated in Section IV.

Algorithm 1: LLM-driven intent-conditioned interactive decision-making framework

Input: Passenger instruction u_k^{pass}, vehicle states S_k, predictions P_k, road and navigation information M_k
Output: Reference trajectory $\Gamma_{0,k}^*$, acceleration $\mathbf{a}_k^*$, steering angle δ_k^{cmd}

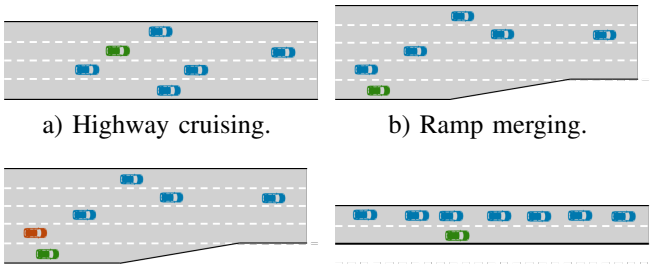
; // Reasoning Module
Initialize history buffer $\mathcal{H} \leftarrow [\,]$ and latest valid result $\mathcal{R} \leftarrow \mathcal{R}^{\text{Normal}}$;
$\mathcal{K}_k^{\text{game}} \leftarrow \text{SelectGame}(S_k, P_k, M_k)$;
$\mathcal{H} \leftarrow \mathcal{H}.\text{Add}(S_k)$;
if $\text{Trigger}(u_k^{\text{pass}}, \mathcal{H})$ *and LLM is idle* **then**
  $\mathcal{K}_k^{\text{LLM}} \leftarrow \text{TopK}(\mathcal{K}_k^{\text{game}}, 3)$;
  $\mathcal{I}_k \leftarrow \mathcal{T}(u_k^{\text{pass}}, S_k, \mathcal{K}_k^{\text{LLM}})$;
  $\text{AsyncQuery}(\mathcal{I}_k)$;
$\mathcal{R} \leftarrow \text{Validate}(\text{ReadLatest}(\mathcal{R}))$;
$(\boldsymbol{\rho}_k, \{\mathbf{p}_{i,k}\}) \leftarrow \mathcal{R}$;
; // Decision Module
$\mathcal{B}_k \leftarrow \text{GenerateBranches}(S_k, M_k, \boldsymbol{\rho}_k)$;
$\mathcal{B}_k^{\text{feas}} \leftarrow \text{Filter}(\mathcal{B}_k, P_k, \boldsymbol{\rho}_k)$;
foreach $b \in \mathcal{B}_k^{\text{feas}}$ **do**
  $\mathbf{a}_b^* \leftarrow \text{SolveEgo}(b, \boldsymbol{\rho}_k)$;
  $\mathcal{A}_b^{\text{SV}} \leftarrow \text{BestResponse}(b, \mathcal{K}_k^{\text{game}}, \Theta)$;
  $J_b^* \leftarrow \text{Evaluate}(b, \mathbf{a}_b^*, \mathcal{A}_b^{\text{SV}}, \{\mathbf{p}_{i,k}\})$;
$b_k^* \leftarrow \arg\max_{b \in \mathcal{B}_k^{\text{feas}}} J_b^*$;
; // Control Module
$(\Gamma_{0,k}^*, \mathbf{a}_k^*) \leftarrow \text{ConstructTrajectory}(b_k^*, \mathbf{a}_{b_k^*}^*)$;
$\delta_k^{\text{cmd}} \leftarrow \text{Stanley}(S_k, \Gamma_{0,k}^*)$;

IV. Experiments and Analysis

This section evaluates whether the proposed method can reliably translate high-level semantic intentions into safe, efficient, and executable driving behavior. We first ablate the surrounding-vehicle type beliefs and the fast–slow coordination mechanism, and then evaluate LLM-based intention reasoning and personalized responses. The overall driving performance is subsequently compared with reinforcement-learning and LLM-driven baselines. Finally, a closed-road vehicle experiment is introduced to examine physical executability.

A. Experimental Setup

1) Simulation Scenarios: The simulation environment is developed on the open-source autonomous-driving platform



a) Highway cruising. b) Ramp merging.

c) Adversarial blocking. d) Dense-traffic cut-in.

Fig. 3. Four simulation scenarios used in the experiments.

highway-env [**?**]. As shown in Fig. 3, the experiments include highway cruising, ramp merging, adversarial blocking, and dense-traffic cut-in scenarios. Background vehicles are divided into Aggressive, Normal, and Cautious types. They share the same IDM/MOBIL structure but use different desired speeds, time headways, standstill gaps, and maximum accelerations. Aggressive vehicles favor higher speeds and shorter gaps, Normal vehicles use intermediate parameters, and Cautious vehicles maintain lower speeds and larger safety margins.

To better simulate a dense cut-in scenario, the ego starts in lane 1 at $x = 200$ m and must merge into lane 0 after the lane-change gate at $x = 230$ m. The key vehicle is the nearest lane-0 vehicle behind the ego: the Cautious key vehicle travels at 6.0 m/s and decelerates at -0.5 m/s^2 to yield when the ego crosses; the Normal key vehicle travels at 8.5 m/s and follows the native IDM; and the Aggressive key vehicle travels at 13.0 m/s with a target speed of 14.0 m/s and accelerates at $+0.5$ m/s^2 to close the gap when the ego's lateral position satisfies $y < 3.5$ m and the longitudinal distance is within 45 m. When the active key vehicle moves more than 5 m ahead of the ego, the nearest lane-0 vehicle within the 100 m relevance range behind the ego is promoted to the key vehicle; the type response of the old key vehicle is removed and restored to native IDM car-following, and the new key vehicle receives its type response.

2) Implementation Details: The slow system uses `qwen-flash-2025-07-28` as the base LLM with a temperature of 0, a 20-frame history at 10 Hz, a 10 s timeout, and no retries. The three nearest game vehicles are selected for prompt construction. The LLM parses the passenger instruction online, produces bounded motion, geometric, and utility parameters, and estimates the type probability of each surrounding vehicle. The fast system includes at most five vehicles in the interaction game, solves the continuous longitudinal action of each candidate path using CasADi/IPOPT, and generates steering commands through a Stanley controller.

The LLM and game-theoretic decision modules operate asynchronously. The slow system updates its parameters and beliefs when the passenger instruction changes or the key traffic state varies significantly, whereas the fast system reads the latest valid output at every planning cycle. If the LLM times out or returns an invalid format, the previous valid parameters are retained. A newly observed vehicle is initialized with the Normal prior.

3) Baselines and Evaluation Metrics: DQN, PPO, DiLu, and THCAD are selected as the main baselines. DQN and PPO represent reinforcement-learning-based decision methods; DiLu uses an LLM for high-level driving reasoning; and THCAD combines an LLM-based slow system with a learning-based fast system. All methods use identical road geometry, traffic initialization, observations, vehicle dynamics, and random seeds.

The task success rate (SR) measures whether the vehicle completes the cruising or merging task without collision or road departure. Average speed measures efficiency, minimum time to collision (Min TTC) measures the safety margin, and acceleration variance measures ride smoothness. For the LLM module, we additionally report classification accuracy, Macro-F1, Brier score, and inference latency. Each scenario is repeated using the same 50 random seeds.

B. LLM Intention Reasoning and Personalized Response

This subsection evaluates the LLM's type beliefs in a closed-loop cut-in ablation, examines the accuracy of its type and preference recognition, and illustrates the resulting behavior in two qualitative cases. We further verify the personalized vehicle response and the online adaptation to a changing passenger instruction.

1) Closed-Loop Ablation of Type Belief: We further isolate the contribution of the surrounding-vehicle type belief to verify the effectiveness of LLM-based type-belief reasoning. Two variants are considered: (1) *Full Model* uses the LLM's soft type probabilities and the complete Keep–Probe–Commit action set; (2) *Uniform Belief* replaces the LLM output with a fixed uniform prior $[1/3, 1/3, 1/3]$, isolating the value of type evidence. The two variants share the scene, seeds, controller, reward, candidate set, and evaluation protocol; only the belief input differs.

The ablation is conducted in the dense-traffic cut-in scenario. Before each run, the system receives the passenger instruction "I'm late for work." The 50 seeds (9950–9999) were frozen before the runs.

The first query is synchronous at step 19; subsequent queries are asynchronous. We report success rate (SR), collisions, average merge time, and average speed. Average merge time is the completion time for successful episodes and the 40 s horizon otherwise. Average speed is the mean over all episodes.

Table I reports the main results. The Full Model completes all 50 episodes without collision, whereas Uniform Belief completes 37/50 (74.0%) and collides in 13. No timeouts occur in either variant. The average merge time is 13.80 s for Full and 19.43 s for Uniform. Average speed is 6.97 m/s for Full and 6.99 m/s for Uniform. This demonstrates that intention inference plays a decisive role in the cut-in maneuver and ensures vehicle safety during the cut-in.

2) Intention-Reasoning Accuracy: For the above ablation runs, we also collect statistics on intention-recognition accuracy.

Table II summarizes recognition over the same 50 seeds; the Full Model succeeds in 50/50 episodes (SR $= 1.00$), and

TABLE I
ABLATION RESULTS IN THE TWO-LANE CUT-IN SCENE OVER 50 RANDOM SEEDS.

Method	SR (%) ↑	Collisions ↓	Average Merge Time (s) ↓	Average Speed (m/s) ↑
Full Model	100.0 (50/50)	0	13.80	6.97
Uniform Belief	74.0 (37/50)	13	19.43	6.99

the recognition metrics are computed from its vehicle–query samples on the same episodes. The three-class task labels each vehicle as Cautious, Normal, or Aggressive, where Aggressive corresponds to the Force intention; Cautious and Normal together form the non-Force class. Across all 1332 vehicle–query samples, three-class accuracy is 70.9%. The 523 key-vehicle samples reach 96.9%, whereas the 809 background-vehicle samples reach 54.0%. The mean latency over all responses is 0.97 s.

Specifically, a query-level sample is defined as one three-class probability vector produced by the LLM for each observed vehicle state in the prompt; a sample is counted as correct when the maximum-probability type matches the true type. The 50 episodes of the Full Model produce 444 responses and 1332 vehicle–query samples, involving 171 vehicle instances.

In this type-observable scenario, the key vehicle exhibits high accuracy: through the interaction between the ego and the key vehicle, 49 of the 50 initial key vehicles are correctly classified at the first observation, and the Force versus non-Force binary classification is 50/50. Over the 444 initial-key queries, the initial key-vehicle query-level accuracy remains at 437/444 (98.4%), consistent with the first-observation accuracy of 98.0%. The key-vehicle query-level Brier score is 0.158, well below the uniform-prediction value of 0.667, indicating that the output is not only correct in the argmax but also concentrates probability mass on the true class.

The low background-vehicle accuracy is likewise structural. On the one hand, background vehicles lack strong interaction with the merging vehicle, so their class-conditional trajectories overlap heavily and the discriminative evidence available to the LLM is weak; even a human would find it difficult to judge the aggressiveness of other vehicles that are not currently interacting. The data show that background-vehicle accuracy is only 48.8% at the first observation and rises to 54.0% after repeated observations. Normal vehicles account for 313/809 (38.7%) of the background samples and constitute the majority-class baseline; the overall accuracy of 54.0% is 15.3 percentage points above this baseline and is therefore reasonable given the weak discriminative evidence. On the other hand, background-vehicle recognition accuracy is less important for closed-loop decisions: the controller injects the observed vehicles' type probabilities into the Bayesian game; because background vehicles are far from the ego and will not intersect the ego's candidate trajectories or conflict corridor in the short term, their type differences have little effect on the game, and recognition errors for distant background vehicles are therefore less important for closed-loop decisions.

In the following two qualitative cases, "without intention inference" corresponds to Uniform Belief and "with intention inference" to the Full Model.

TABLE II
LLM INTENTION-REASONING AND VEHICLE-TYPE RECOGNITION RESULTS.

Task / subset	Query samples	Accuracy (%)	Macro-F1	Brier	Force vs non-Force (%)
Cut-in, all vehicles	1332	70.9	0.702	0.427	84.6
Cut-in, key vehicle	523	96.9	0.960	0.158	99.8
Cut-in, background vehicles	809	54.0	0.533	0.601	74.8
Cut-in, key vehicle (first obs.)	50	98.0	0.982	0.154	100.0

3) Adversarial Blocking Case: To visualize the effect of probabilistic intention inference, we compare two runs with identical initial states. The yellow vehicle is the AV, and the red vehicle is the interacting vehicle. Without intention inference, the AV assumes a homogeneous Aggressive response and accelerates to overtake at $t = 3$ s, as shown in Fig. 4(a). The interacting vehicle then accelerates to block the merge. Because its adversarial intention is not represented, both vehicles continue competing for the gap, forcing the AV to brake near the end of the merging lane at $t = 10$ s.

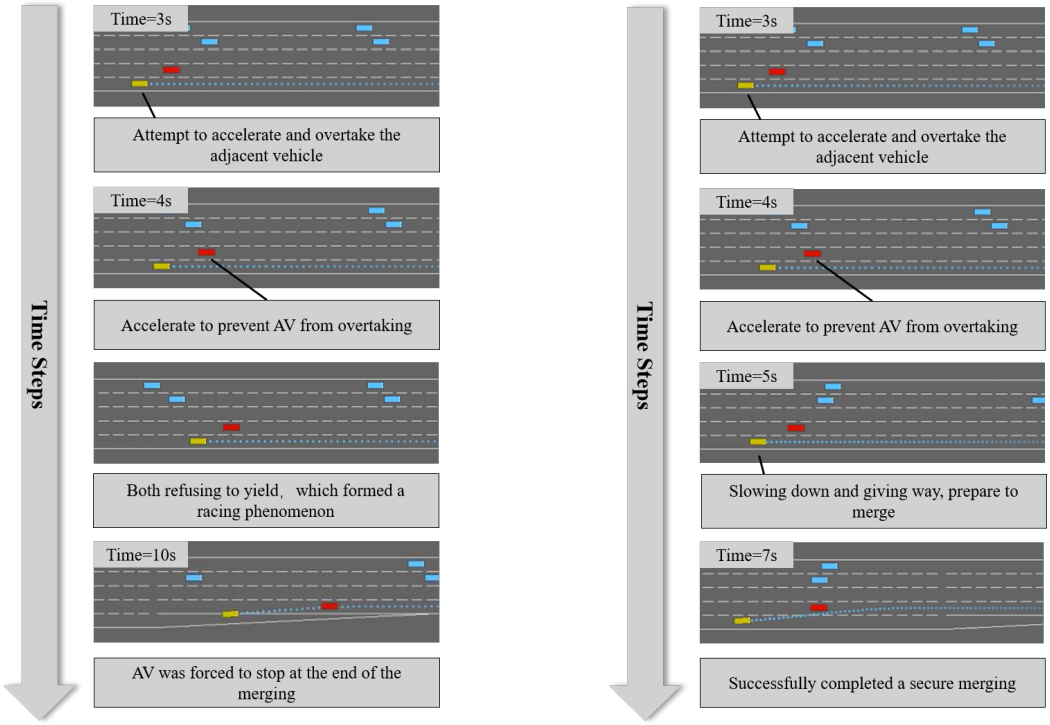
In contrast, the full model assigns a higher probability to the Force type after observing the blocking behavior. The AV abandons the overtake, decelerates at $t = 5$ s, and completes a safe merge at $t = 7$ s, as shown in Fig. 4(b). This case demonstrates that probabilistic intention inference converts an emerging acceleration race into proactive yielding before the safety fallback is activated.

4) Dense-Traffic Cut-In Case: To analyze the effect of intention inference on gap selection, we compare two runs with identical initial states. The yellow AV must merge into the congested target lane through a short and dynamically changing gap. Without intention inference, the AV approaches the lane boundary but cannot interpret the adjacent vehicle's deceleration as yielding, as shown in Fig. 5(a). It therefore abandons the maneuver and returns to its original lane. The following vehicle then closes the gap, leaving the AV trapped in a repeated probing-and-retreating process and eventually causing a collision at $t = 12$ s.

With intention inference, the observed deceleration increases the belief that the adjacent vehicle is of the Yield type. The AV consequently commits to the cut-in at $t = 3$ s and completes the merge at $t = 4$ s, as shown in Fig. 5(b). This case demonstrates that probabilistic intention inference converts a subtle yielding response into an actionable merging opportunity and avoids indecisive behavior in dense traffic.

These cases provide qualitative interpretations of the module behavior; the final conclusions are based on the multi-seed statistics in Table I.

5) Personalized Vehicle Response: To verify whether natural-language instructions affect vehicle behavior through model parameters, we pair the urgent instruction "I am late



a) Without intention inference. b) With probabilistic intention inference.

Fig. 4. Behavioral comparison in the adversarial blocking scenario.

for work" with the comfortable instruction "I still have ample time" under identical traffic flows and random seeds. The LLM parses each instruction online and outputs bounded motion, geometric, and utility parameters. In addition to validating their ranges, we compare average speed, acceleration variance, and Min TTC to examine the complete transmission chain from language instruction to model parameters, game decision, and vehicle trajectory.

In Fig. 6, the upper and lower rows correspond to strongly interactive and highway-cruising scenarios, respectively. The left column compares speed distributions, the middle column shows the joint efficiency-comfort distribution, and the right column presents the cumulative Min TTC distribution. The current results indicate that the urgent preference generally produces higher speeds, whereas the comfortable preference yields lower acceleration variance. Because the urgent preference also produces a slightly lower Min TTC in some scenarios, its cost must be assessed jointly using collision rate, safety thresholds, and significance tests rather than equating personalization with speed improvement.

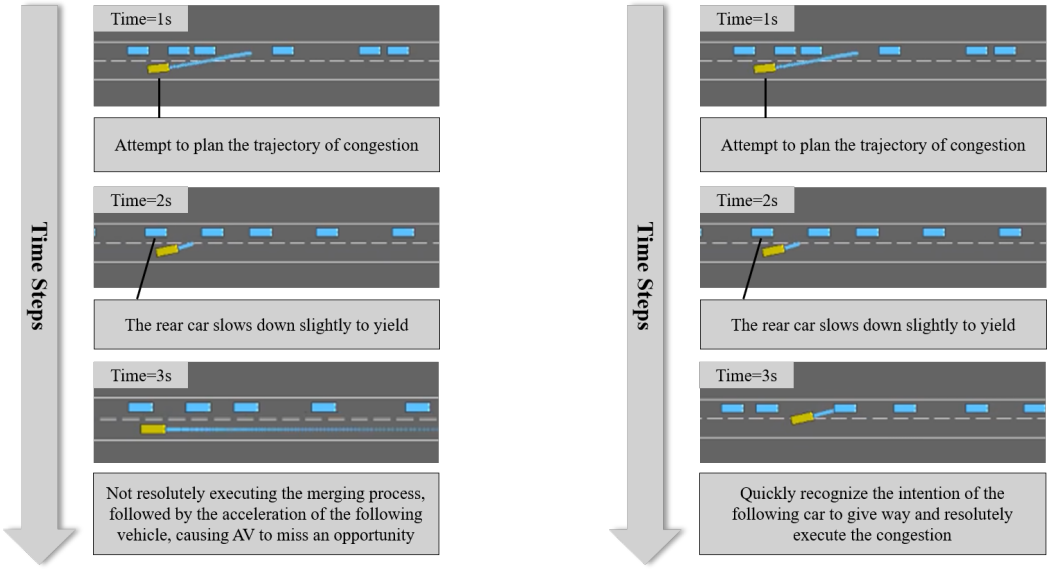
6) Online Passenger-Intent Adaptation Case: To illustrate the online response to changing passenger intent, we conduct a cruising experiment with random seed $S = 9$. The initial instruction, "I still have ample time," is interpreted as a conservative preference. By $t = 7$ s, the AV has established stable car-following with a relatively large gap and does not overtake the faster vehicle in the adjacent lane, as shown in Fig. 7.

At $t = 19$ s, the instruction changes to "I'm late for work." The slow system updates the style and motion parameters toward higher efficiency, after which the fast system initiates a left lane change and accelerates. By $t = 28$ s, the AV has established a new steady following state at a higher speed and with a shorter gap. The complete transition is shown in Fig. 7. This case demonstrates that the proposed framework can translate an online semantic preference change into consistent longitudinal and lateral behavior without interrupting the planning process.

C. Comparison with Other Methods

1) Overall Performance: We first compare the task success rates in highway-cruising and ramp-merging scenarios. As shown in Fig. 8, the current results indicate that PPO, THCAD, and our method achieve a 100% success rate in highway cruising. In the more interactive ramp-merging scenario, our method maintains a 100% success rate, whereas DiLu, DQN, PPO, and THCAD achieve 46%, 64%, 70%, and 84%, respectively. These results suggest that explicitly modeling surrounding responses and uncertain type beliefs improves stability in complex merging interactions. All values will be reproduced using the final implementation and unified random seeds before submission.

Table III further reports average speed, acceleration variance, and Min TTC. Unlike baselines that primarily pursue speed or adopt a uniformly conservative strategy, our method adjusts efficiency and comfort through passenger-preference parameters and constrains strong interactions using probabilistic type beliefs. The current data show high average speeds and low acceleration variance in both scenarios. Although



a) Without intention inference. b) With intention inference.

Fig. 5. Behavioral comparison in the dense-traffic cut-in scenario.

TABLE III
SIMULATION RESULTS OF DIFFERENT METHODS.

Scenario	Method	SR (%) ↑	Average Speed (m/s) ↑	Acceleration Variance (m^2/s^4) ↓	Min TTC (s) ↑
Highway	DiLu	78†	23.30†	0.43†	23.92†
	DQN	48†	24.52†	6.17†	7.45†
	PPO	100†	22.47†	0.41†	31.88†
	THCAD	100†	25.46†	0.38†	8.65†
	Ours–Urgent	100†	27.76†	0.1199†	9.32†
	Ours–Comfort	100†	27.30†	0.0964†	10.54†
Merge	DiLu	46†	15.88†	2.01†	7.59†
	DQN	64†	18.69†	3.72†	2.62†
	PPO	70†	16.31†	1.96†	5.21†
	THCAD	84†	14.51†	0.32†	18.93†
	Ours–Urgent	100†	23.16†	0.9704†	10.32†
	Ours–Comfort	100†	22.81†	0.9291†	10.55†

†Results from the previous thesis implementation; all values will be rerun and updated using the final code.

the Min TTC is lower than that of some highly conservative configurations, the task success rate is not reduced. Interactive-driving methods should therefore be evaluated jointly in terms of success, safety margin, efficiency, and comfort.

2) Safety, Efficiency, and Comfort: PPO improves system safety by maximizing the minimum time to collision (Min TTC). However, its strong emphasis on safety reduces travel efficiency. In contrast, the DQN-based method primarily pursues traffic efficiency but exhibits large acceleration fluctuations and is prone to rear-end conflicts, particularly in congested conditions, indicating an insufficient balance between efficiency and safety. DiLu can effectively understand and execute driving instructions, but its performance deteriorates as traffic density increases, resulting in a lower task success rate and discontinuous planned trajectories. THCAD adopts a relatively low average speed of approximately $14.51\,\mathrm{m/s}$ in merging scenarios to improve safety, but this strategy is overly conservative and exhibits risk-avoidance behavior.

Under the instruction "I am late for work," the proposed method achieves an average speed of $27.76\,\mathrm{m/s}$ while limiting the acceleration variance to $0.1199\,\mathrm{m^2/s^4}$, substantially lower than those of PPO ($0.41\,\mathrm{m^2/s^4}$) and DQN ($6.17\,\mathrm{m^2/s^4}$). These results indicate that the proposed method maintains smooth vehicle motion while achieving efficient and collision-free driving. Despite its higher travel speed, it retains a Min TTC of approximately $9.32\,\mathrm{s}$, which remains well above the collision-risk threshold. Although DQN also achieves a relatively high average speed of $24.52\,\mathrm{m/s}$, its substantially larger acceleration variance degrades ride comfort.

The advantage of the proposed method is more evident

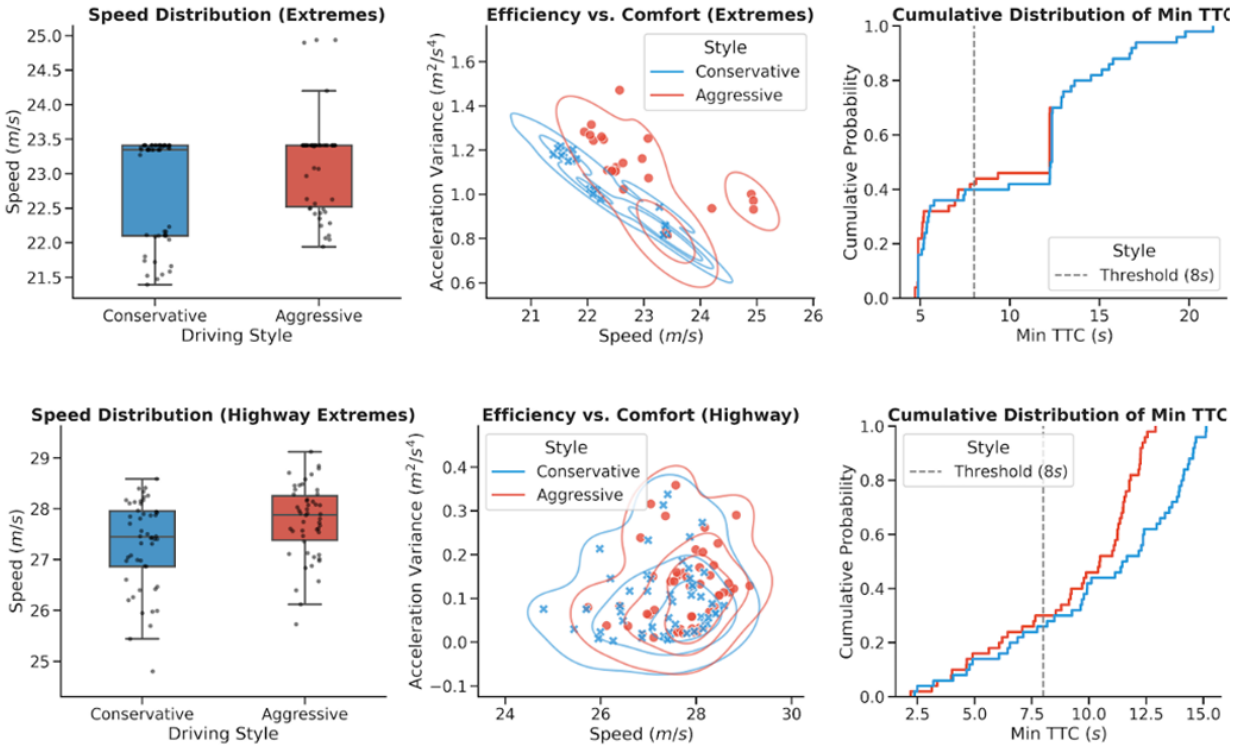


Fig. 6. Statistical analysis of ego-vehicle responses under different passenger-preference instructions.

in the challenging merging scenario, where it achieves an average speed of $23.16\,\mathrm{m/s}$. This result demonstrates that the Bayesian game framework enables the ego vehicle to exploit fine-grained interactive behaviors and improve merging efficiency while maintaining safety.

D. Field Testing and Real-Vehicle Validation

To examine the physical executability of the proposed intent-conditioned fast–slow decision-making framework, field experiments were conducted at Tongji Small Town (TJST) in Shanghai, China. The experiments focused on interactive lane changing along a straight-road section of the test site. An autonomous vehicle (AV) interacted with a human-driven vehicle (HV) traveling in the adjacent target lane. Two representative cases were considered: the HV yielded to facilitate the lane change, or accelerated to maintain its priority. These cases examined whether the AV could adapt its merging strategy to different human responses.

The vehicle experiments followed the proposed fast–slow architecture. Structured descriptions of the traffic scene provided the slow system with information for surrounding-vehicle intention reasoning. The resulting probabilistic type beliefs parameterized the Bayesian–Stackelberg interaction game, through which the fast system evaluated candidate maneuvers and determined the corresponding longitudinal actions. The selected trajectory was then passed to the vehicle control layer for execution. This architecture connected semantic intention reasoning with executable lateral and longitudinal behavior, while allowing the fast decision layer to operate between successive slow-system updates.

Fig. 9 presents two representative interaction sequences. In both cases, the AV initially traveled alongside the HV and intended to enter its lane. The AV subsequently approached the target lane, making its lane-change intention observable through its lateral motion. The HV's response then provided behavioral evidence for distinguishing a yielding interaction from a competing one.

In the yielding case shown in the left column of Fig. 9, the HV slightly reduced its speed and allowed the AV to proceed first. This response was consistent with the *Yield* type defined in the proposed framework and created an opportunity for the AV to merge ahead. The AV proceeded with the lane change and completed the maneuver in front of the HV. This sequence illustrates how a yielding response can support a decisive transition from approaching the target lane to completing the merge.

In the competing case shown in the right column of Fig. 9, the HV accelerated instead of yielding, reducing the opportunity for the AV to merge ahead. This behavior was consistent with the *Force* response model, in which the surrounding vehicle prioritizes its own progress. The AV responded by reducing its speed and allowing the HV to pass, before completing the lane change behind it. The resulting maneuver illustrates an alternative interaction strategy in which the AV adjusts the passing order to accommodate the surrounding driver's behavior.

Together, these field cases provide qualitative evidence

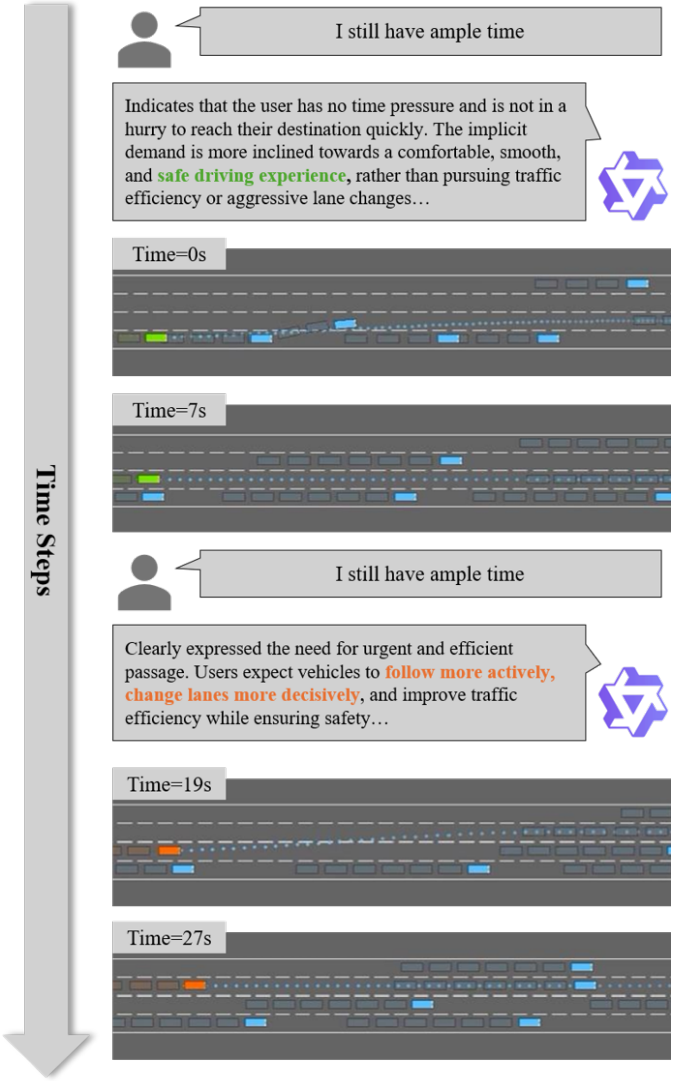


Fig. 7. Online response to a change in passenger intent from a time-insensitive request to an urgent travel request.

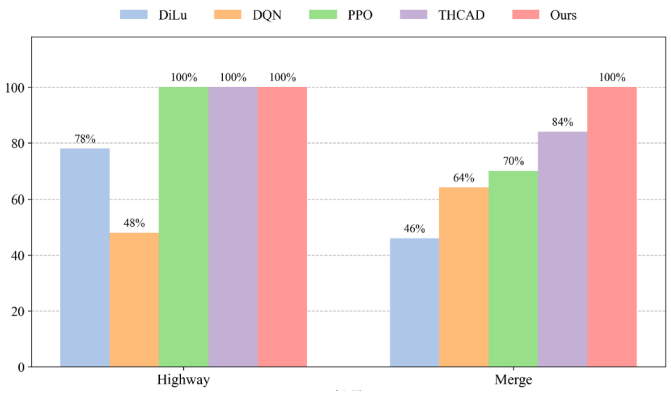


Fig. 8. Task success rates in highway-cruising and ramp-merging scenarios.

of the framework's physical executability in interactive lane changing. The AV completed the same lateral task through different longitudinal strategies, merging ahead of a yielding HV and behind a competing HV. These observations complement the simulation results by demonstrating adaptive vehicle behavior during interactions with a human driver.

REFERENCES

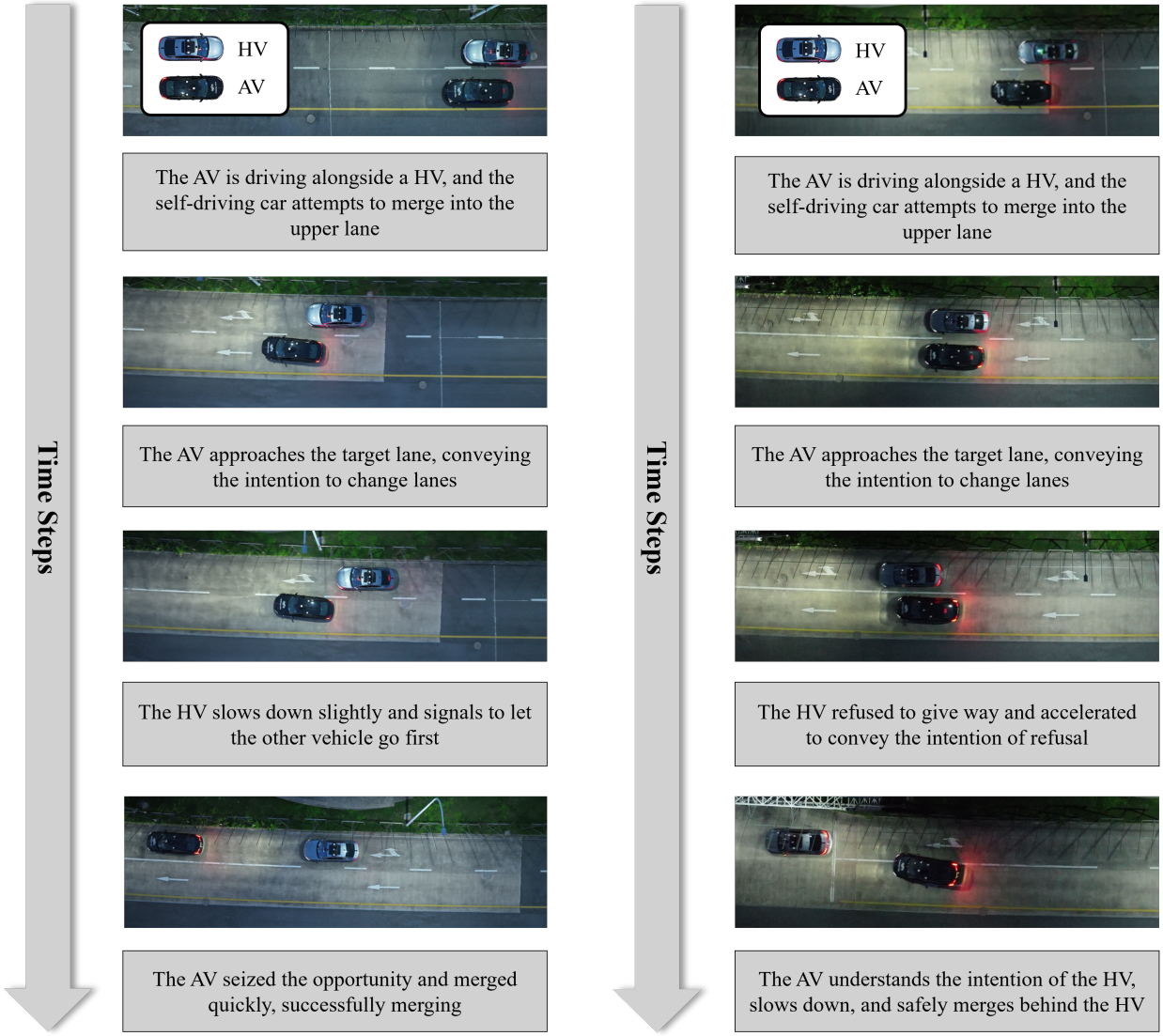


Fig. 9. Representative real-vehicle lane-changing interactions at Tongji Small Town. Left: the HV yields and the AV merges ahead. Right: the HV accelerates and the AV merges behind. Time progresses from top to bottom in each column.